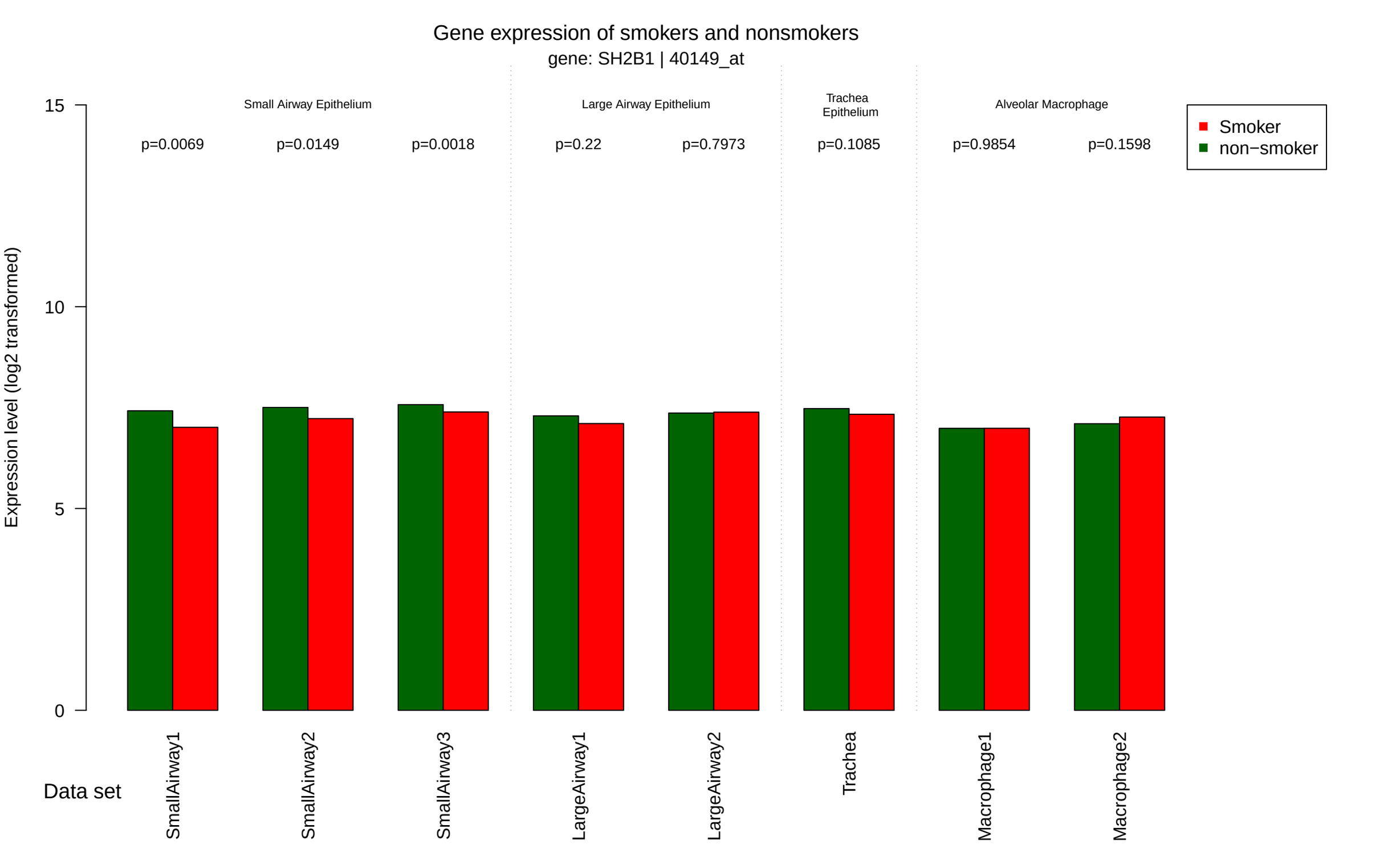
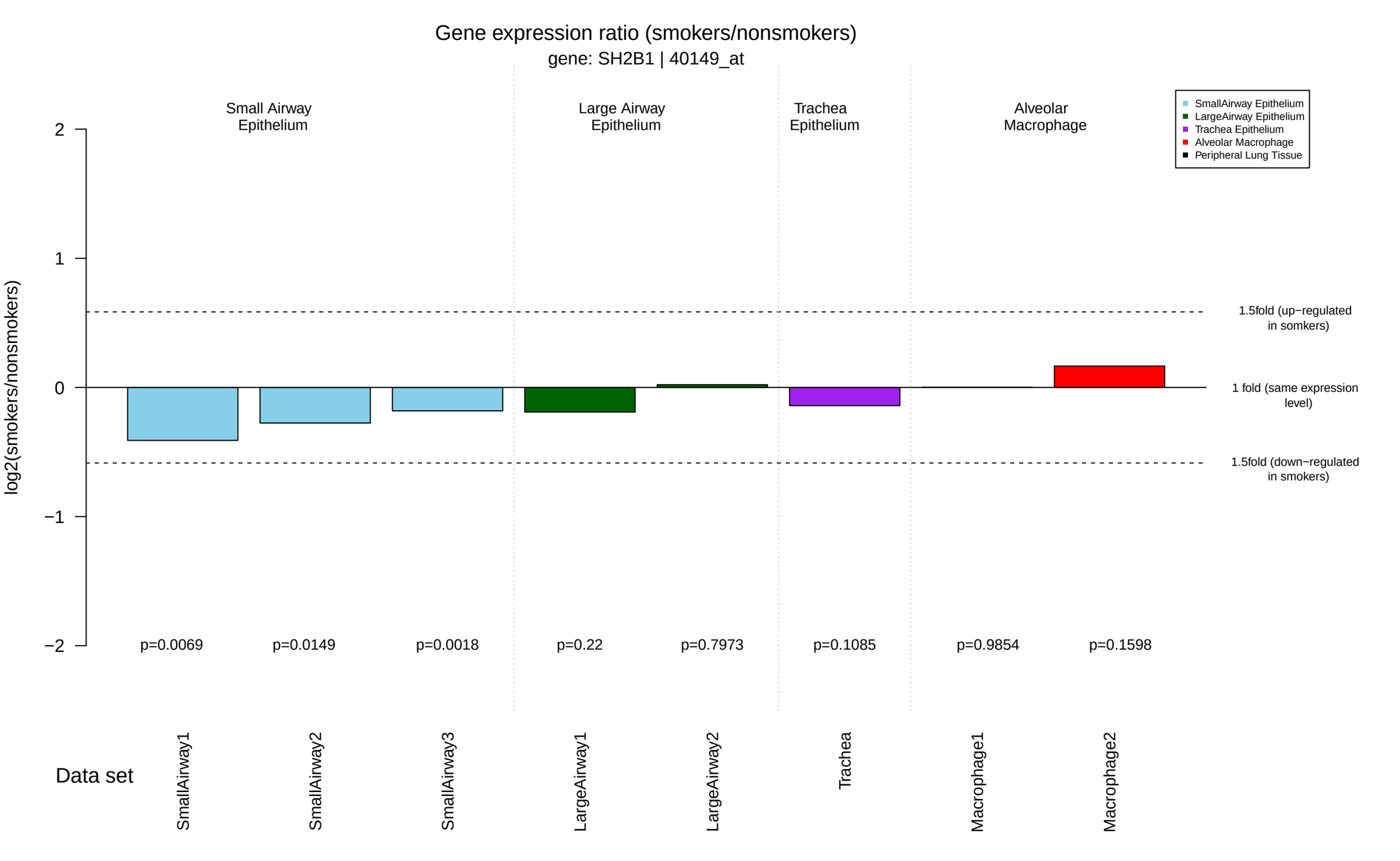
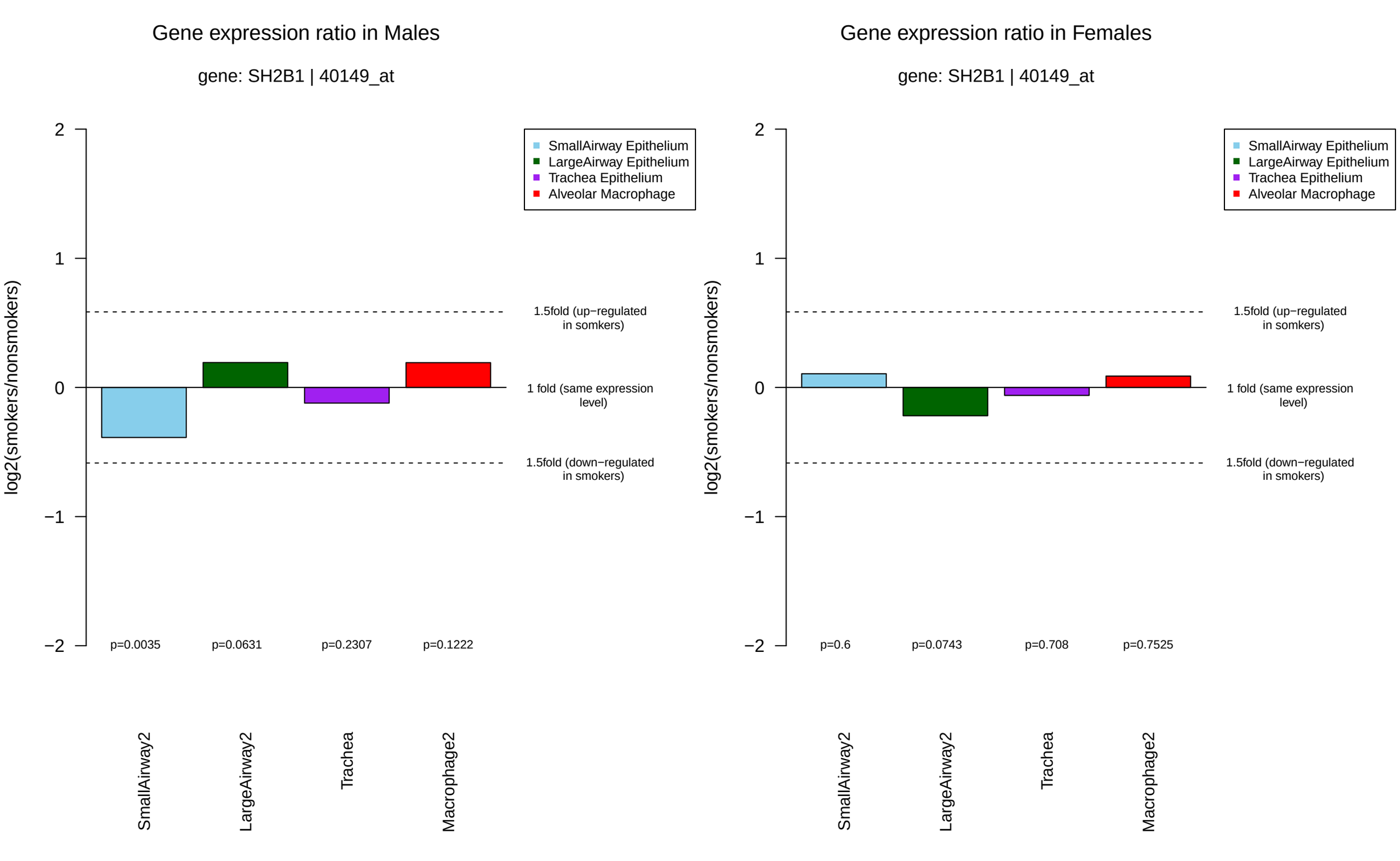
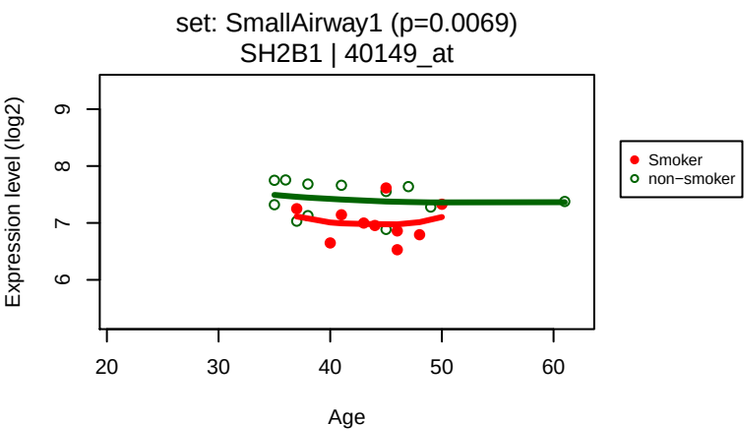
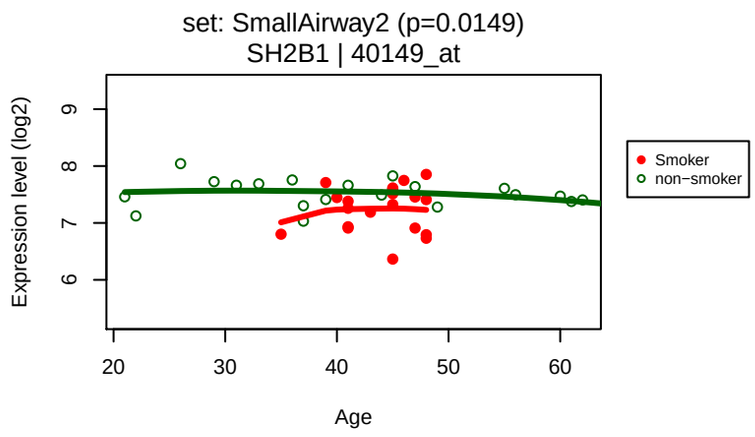
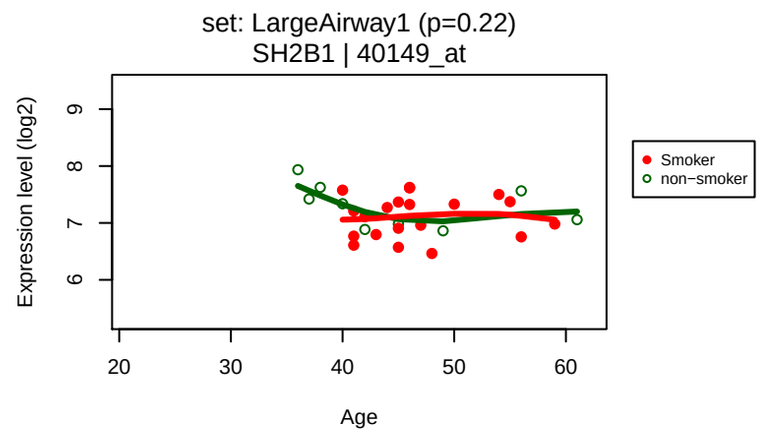

Gene expression of smokers and nonsmokers
gene: SH2B1 | 40149_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0069
p=0.0149
p=0.0018
p=0.22
p=0.7973
p=0.1085
p=0.9854
p=0.1598
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

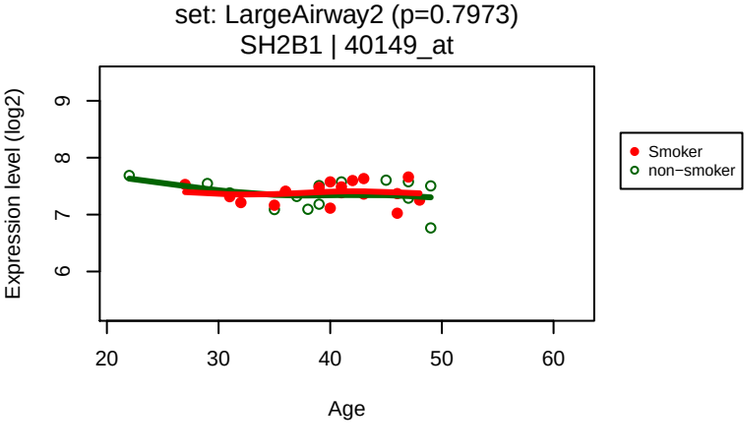

Gene expression ratio (smokers/nonsmokers)
gene: SH2B1 | 40149_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0069
p=0.0149
p=0.0018
p=0.22
p=0.7973
p=0.1085
p=0.9854
p=0.1598
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

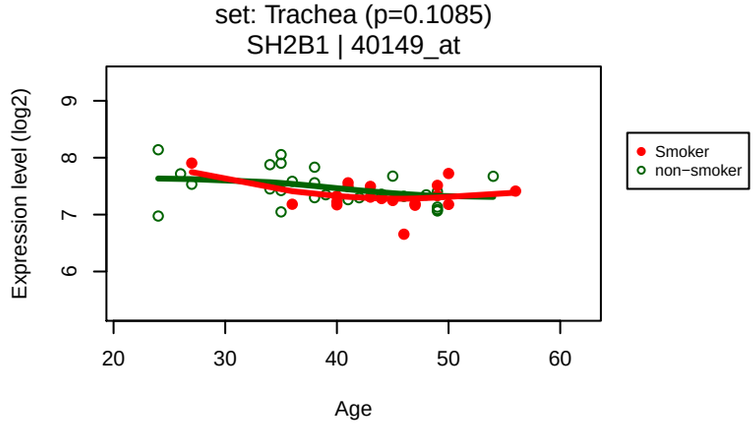

Gene expression ratio in Males
gene: SH2B1 | 40149_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0035
p=0.0631
p=0.2307
p=0.1222
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SH2B1 | 40149_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6
p=0.0743
p=0.708
p=0.7525
SmallAirway2
LargeAirway2
Trachea
Macrophage2

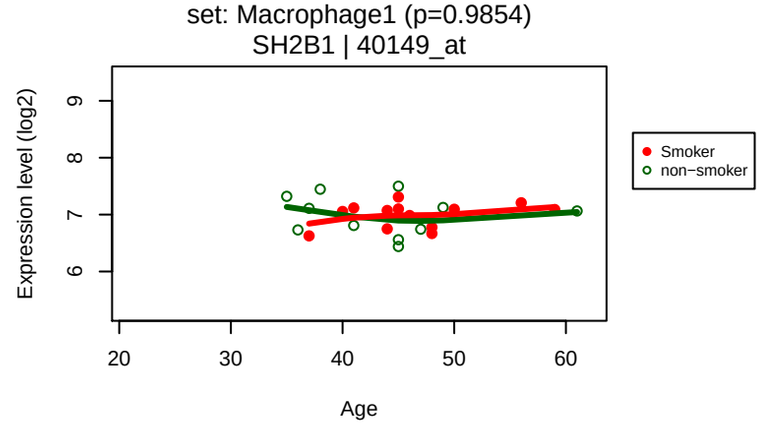

set: SmallAirway1 (p=0.0069)
SH2B1 | 40149_at
Expression level (log2)
Age
Smoker
non−smoker

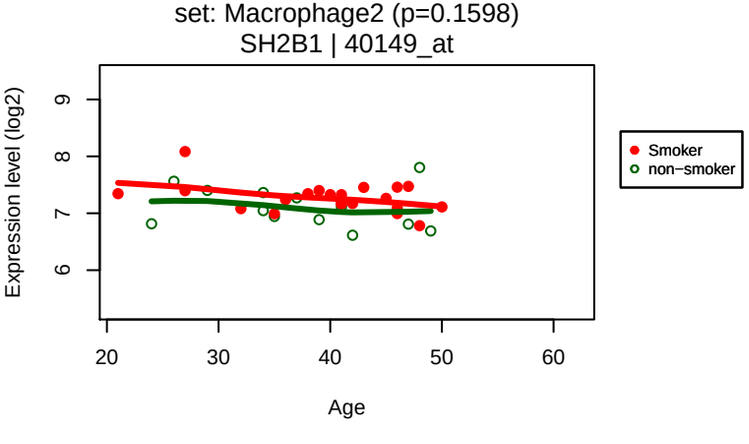

set: SmallAirway2 (p=0.0149)
SH2B1 | 40149_at
Expression level (log2)
Age
Smoker
non−smoker

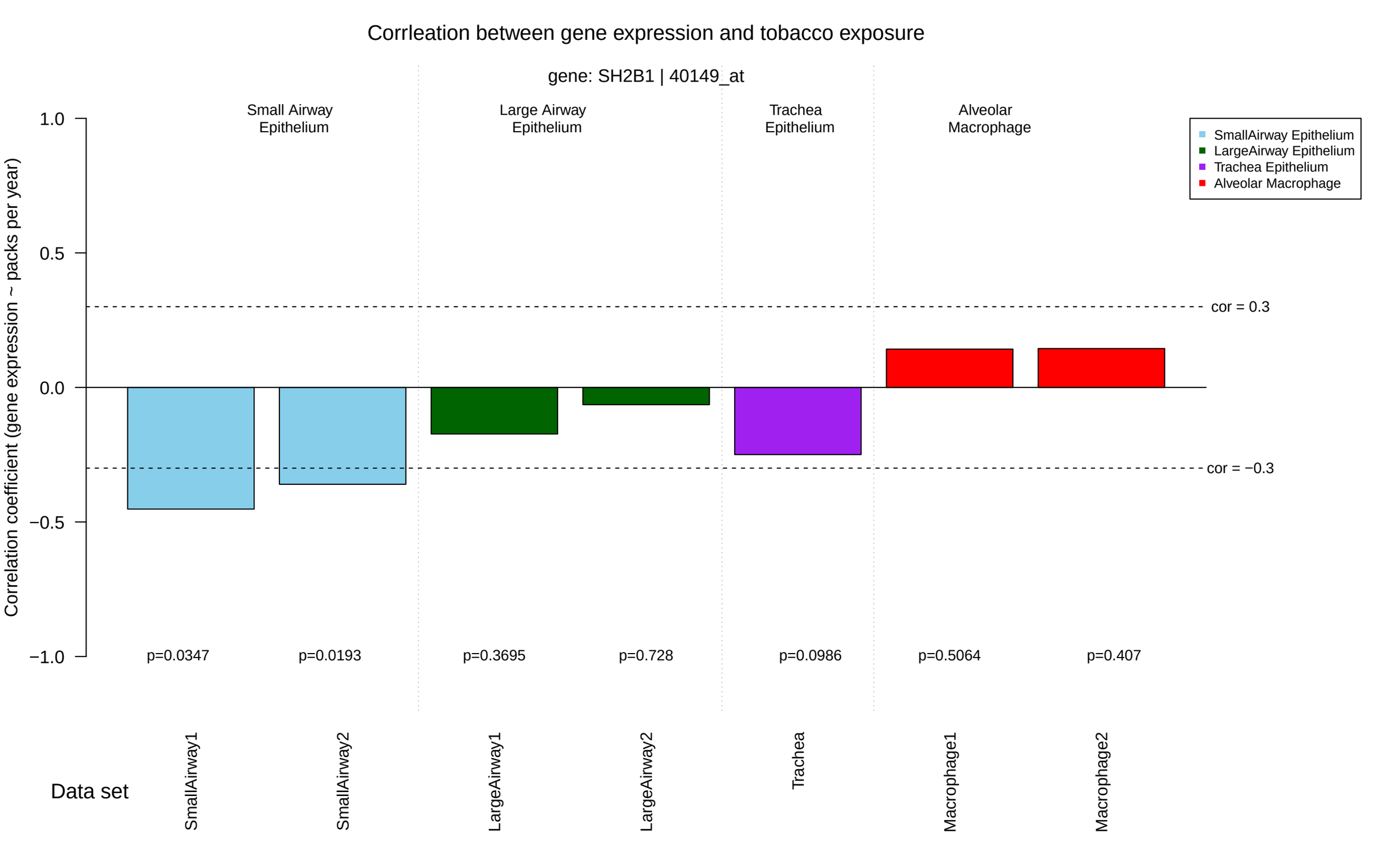

set: LargeAirway1 (p=0.22)
SH2B1 | 40149_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7973)
SH2B1 | 40149_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1085)
SH2B1 | 40149_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9854)
SH2B1 | 40149_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1598)
SH2B1 | 40149_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SH2B1 | 40149_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0347
p=0.0193
p=0.3695
p=0.728
p=0.0986
p=0.5064
p=0.407
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2